

EE3621: Digital Signal Processing

Lecture 24: FIR Design via Frequency-Sampling Method (III B.Tech EEE)

September 9, 2026

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Sample** desired frequency responses $H_d(e^{j\omega})$ with proper linear-phase phase assignments.
 2. **Compute** FIR filter coefficients $h[n]$ via IDFT formulation.
 3. **Optimize** transition band samples to maximize stopband attenuation.
 4. **Compare** the Frequency-Sampling method with the Windowing method.
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2. MATHEMATICAL FOUNDATIONS

2.1 Frequency-Sampling Design Equations

Let $H[k] = H_d(e^{j\frac{2\pi k}{N}})$. For a linear-phase FIR filter of length N :

$$H[k] = |H[k]|e^{j\angle H[k]}, \quad \angle H[k] = -\left(\frac{N-1}{2}\right)\frac{2\pi k}{N}$$

The impulse response is computed by the IDFT:

$$h[n] = \frac{1}{N} \sum_{k=0}^{N-1} H[k] e^{j\frac{2\pi kn}{N}}$$

Exploiting Hermitian symmetry ($H[N-k] = H^*[k]$):

- **For N Odd:**

$$h[n] = \frac{H[0]}{N} + \frac{2}{N} \sum_{k=1}^{(N-1)/2} |H[k]| \cos \left[\frac{2\pi k}{N} \left(n - \frac{N-1}{2} \right) \right]$$

- **For N Even:**

$$h[n] = \frac{H[0]}{N} + \frac{2}{N} \sum_{k=1}^{N/2-1} |H[k]| \cos \left[\frac{2\pi k}{N} \left(n - \frac{N-1}{2} \right) \right] + \frac{H[N/2]}{N} (-1)^n$$

2.2 Transition Sample Optimization Values

— Number of Transition Samples — Optimized Sample Values — Resulting Stopband Attenuation A_s — — $\vdash$ — $\vdash$ — $\vdash$ — — **0 (No transition band)** — None ($H[k] \in \{0, 1\}$) — —16 dB — — **1 Transition Sample** — $T_1 \approx 0.38$ — -45 dB — — **2 Transition Samples** — $T_1 \approx 0.59$; $T_2 \approx 0.11$ — -75 dB — — **3 Transition Samples** — $T_1 \approx 0.70$; $T_2 \approx 0.25$; $T_3 \approx 0.02$ — -95 dB —

Pictorial Representations

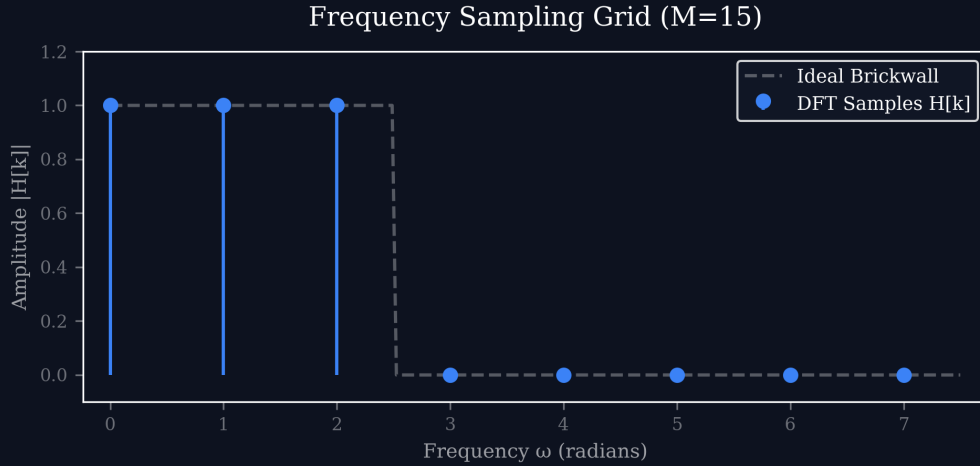


Figure 1: Freq Sampling Discrete

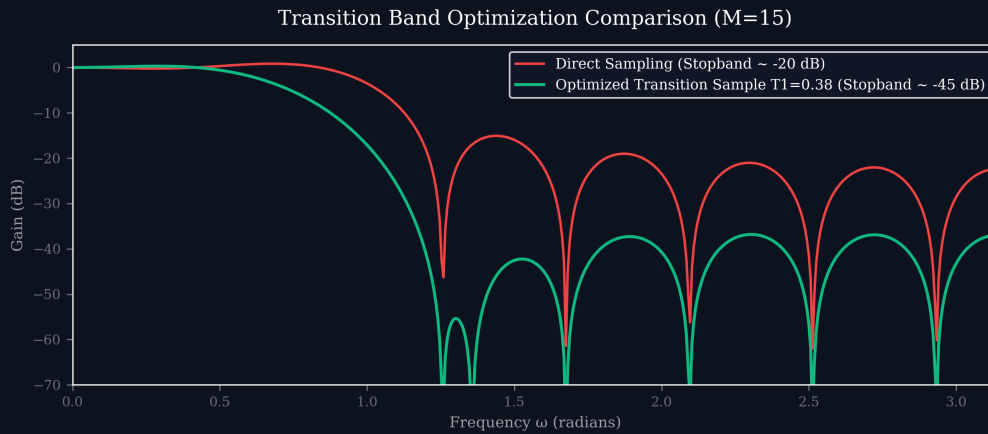


Figure 2: Transition Samples

3. WORKED NUMERICAL EXAMPLES

Example 24.1: Frequency-Sampling Lowpass Filter Design

Problem: Design a 7-point ($N = 7$) linear-phase FIR lowpass filter with frequency samples:

$$|H[k]| = 1, 1, 0, 0, 0, 0, 1 \quad \text{for } k = 0, 1, 2, 3, 4, 5, 6$$

Solution:

- Length $N = 7$ (Odd), $\tau = \frac{7-1}{2} = 3$.
- Non-zero samples: $|H[0]| = 1$; $|H[1]| = |H[6]| = 1$.
- Using the real summation formula:

$$h[n] = \frac{H[0]}{7} + \frac{2}{7}|H[1]| \cos \left[\frac{2\pi(1)}{7}(n-3) \right] = \frac{1}{7} + \frac{2}{7} \cos \left[\frac{2\pi}{7}(n-3) \right]$$

$$\begin{aligned} * h[3] &= \frac{1}{7} + \frac{2}{7} \cos(0) = \frac{3}{7} \approx 0.4286 & * h[2] &= h[4] = \frac{1}{7} + \frac{2}{7} \cos\left(\frac{2\pi}{7}\right) = \frac{1+2(0.6235)}{7} = \frac{2.2470}{7} \approx 0.3210 \\ * h[1] &= h[5] = \frac{1}{7} + \frac{2}{7} \cos\left(\frac{4\pi}{7}\right) = \frac{1+2(-0.2225)}{7} = \frac{0.5550}{7} \approx 0.0793 & * h[0] &= h[6] = \frac{1}{7} + \frac{2}{7} \cos\left(\frac{6\pi}{7}\right) = \\ &= \frac{1+2(-0.9010)}{7} = \frac{-0.8019}{7} \approx -0.1146 \end{aligned}$$

$$h[n] = -0.1146, 0.0793, 0.3210, 0.4286, 0.3210, 0.0793, -0.1146$$

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4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) Derive the Frequency-Sampling design formula for an odd-length linear-phase FIR filter. *(8 Marks)* (b) Explain why transition band samples are introduced in the Frequency-Sampling method. How do they affect the passband ripple and stopband attenuation? *(7 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**

* Mathematical derivation using IDFT and Euler's formula *(8 Marks)*

- **Part (b):**

* Explanation of interpolation ripple between sharp 1-to-0 transitions *(3 Marks)* * Role of transition samples T_1, T_2 in smoothing frequency gradient and boosting A_s from 16 dB to > 45 dB *(4 Marks)*

5. PYTHON VERIFICATION SCRIPT

```
import numpy as np
```

```
N = 7
```

```
H = np.array([1, 1, 0, 0, 0, 0, 1], dtype=complex)
```

```
\title{\textbf{EE3621: Digital Signal Processing}} \Large Apply linear phase \\\large (I
```

```
k = np.arange(N)
```

```
H_phase = H * np.exp(-1j * ((N - 1) / 2) * 2 * np.pi * k / N)
```

```
h = np.fft.ifft(H_phase).real
```

```
print("Frequency Sampling Coefficients h[n]:", np.round(h, 4))
```